



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

Flow

- Announcement
- Recap
- Introduction

Announcement

- Lectures
 - Change of venue for classes: BSB 203
 - M 8–8:50 am; T 1–1:50 pm; R 11–11:50 am; F 10–10:50 am
 - Fridays usually (*but not always*) off
- Tutorials (10-12 in number)
 - Monday: 3–5 pm
 - Venue: **to be decided**
- Check Moodle regularly (courses.iitm.ac.in)
 - *Course outline will be uploaded later today*

Messages from the Last Class

Message 1: Physics vs. Engineering

- Physics: What is true?
- Engineering: what to do? (given our observations & models are imperfect)
- Real world $\rightarrow$ OUR OBSERVATIONS $\rightarrow$
OUR MODELS $\rightarrow$ PREDICTIONS $\rightarrow$

Decisions

Message 2: All problems can be reduced to 3 layers

- Layer 1: what we know (laws, observations, PAST)
- Layer 2: what we don't know (actual event, FUTURE
Some of PAST)
- Layer 3: what we must do (design, punish/acquit, PRESENT)

Message 3: Hypothesis Testing

What is this course about?

Yesterday's Question

- Engineers must take decisions
"Before certainty arrives".

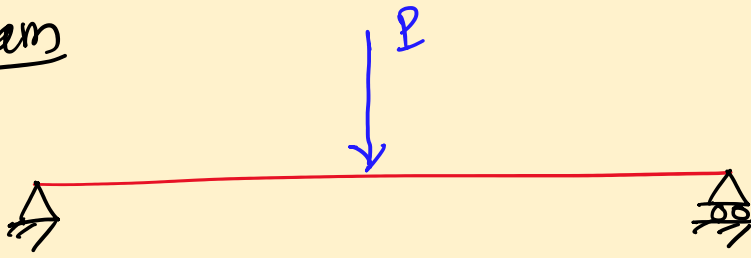
Prior
events
Posterior



- How do we make rational decisions?

Deterministic Delusion

Beam



increase P until the beam fails @ say, P_0

P_0 varies because of

- * variable material
- * measurement of error
- * variable load

→ What is the failure load?

True Strength



Test
Cubes
(150 mm)



observed failure load
→ 15, 17, 13, 14, 12, 20, 11
(11-20)

mean, median, minimum, $\mu - \sigma$,
 μ

Factor of Safety

Objective of Design

Resistance $>$ Load
in a deterministic world.

Uncertain Resistance $>$ Uncertain Load
in a probabilistic world.

- When uncertainty is low & insignificant consequence
 - Use mean, e.g., your height, dimension, E
- When uncertainty is large
 - Play safe
 - Factor of Safety (FoS) compensates for uncertainty.

Margin of Safety

Let's define Margin of Safety

$$Z = R - S$$

R: Resistance (Resister) Capacity

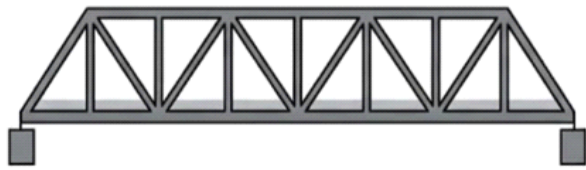
S: Stress (Solicitation) Demand

When $Z > 0 \Rightarrow$ SAFE

When $Z < 0 \Rightarrow$ UNSAFE

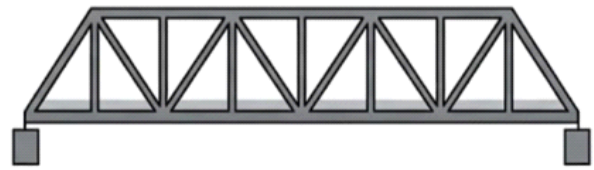
Which one is safer?

BRIDGE A



- Mean Capacity = 500 kN
- Mean Load = 300 kN
- FoS = 1.67

BRIDGE B



- Mean Capacity = 500 kN
- Mean Load = 300 kN
- FoS = 1.67

Capacity: Almost constant
Demand: Almost constant

Highly variable
Highly variable

→ Bridge A is safer.

→ WHY?

~~BRIDGE IS SAFE.~~

Bridge has acceptably low
probability of failure.

Risk is more than Probability

Probability $\times$ Consequence $\rightarrow$ Risk

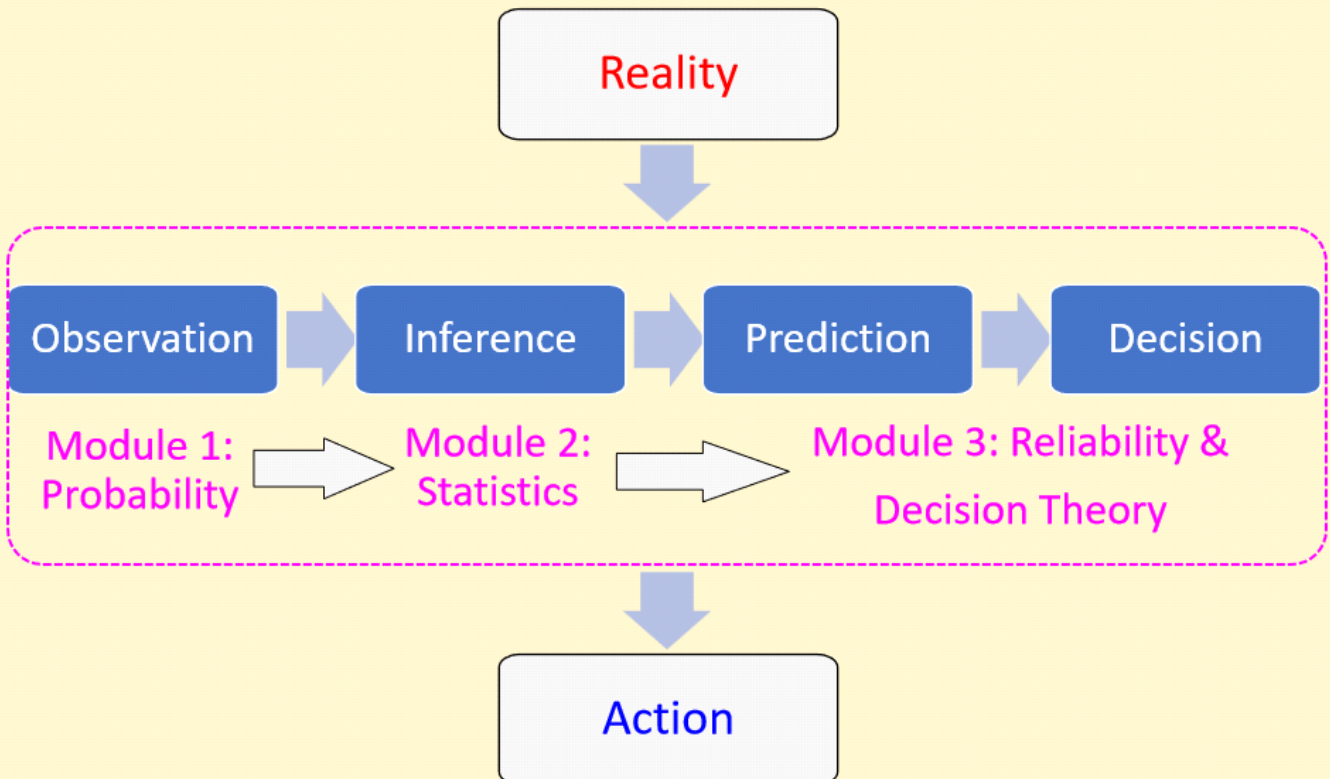
EARTHQUAKES : Himalayas: Pr is high.

Chennai: Pr is low.

CONSEQUENCES : Pr that building collapses
given a fixed earthquake

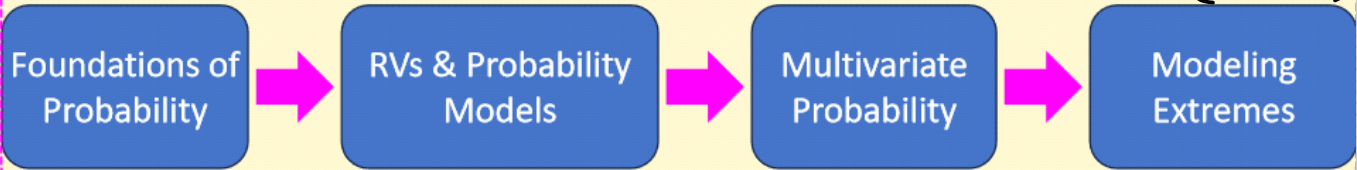
Himalayas: Low
Chennai: high

Where are we going?



Course Flow

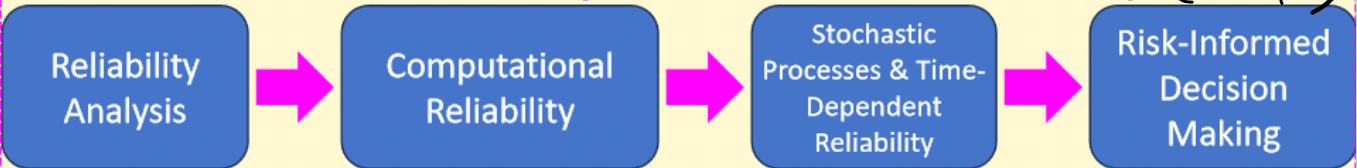
Module 1: **Probability** – Understanding Uncertainty (16–18)



Module 2: **Statistics** – Learning from Data (12–15)



Module 3: **Reliability** – Decision under Uncertainty (10–12)



Next Class

- How do we mathematically describe uncertainty?
 - Sample space
 - Events
 - Set algebra
- Suggested reading:
 - Benjamin-Cornell, Ch. 2.1
 - Papoulis, Ch. 2.1
 - Ang & Tang, Vol I, Ch. 2.

Thank you!